

SIVA D

SR SOFTWARE ENGINEER

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|LinkedIn | GitHub | Portfolio



Professional Summary:

Sr Software Engineer with around 8 years of experience designing, optimizing, and deploying high-performance, scalable applications. Proficient in Java ecosystem, with expertise in cloud-native architectures and data-driven solutions. Skilled in crafting end-to-end web applications using React, Angular, and REST/GraphQL APIs. Committed to integrating machine learning for innovative software solutions, ensuring code excellence through CI/CD pipelines, and thriving in collaborative Agile environments.

- Developed robust multi-tier web applications using Java and Scala, leveraging Spring Boot for REST APIs, security, cloud integration, and batch processing.
- Designed and implemented microservices architectures to ensure scalability and maintainability of enterprise applications.
- Built responsive and interactive user interfaces using Angular, React, JavaScript, TypeScript, HTML5, CSS3, and RxJS, enhanced with Bootstrap for mobile-first designs.
- Configured and managed application and web servers, including Apache Tomcat, JBoss, and Nginx, for optimal performance and deployment.
- Created scalable RESTful APIs and GraphQL endpoints with Spring Boot and JAX-RS, incorporating input validation, exception handling, caching, and comprehensive API documentation via Swagger.
- Engineered real-time, event-driven systems using Apache Kafka and ActiveMQ to enable seamless asynchronous service communication.
- Deployed and managed containerized applications using Docker and Kubernetes, integrating CI/CD pipelines with Jenkins and GitHub Actions for automated workflows.
- Leveraged cloud platforms like AWS (Lambda, S3, EC2, CloudWatch, IAM) and Azure (AKS, Azure Functions, App Services, Azure SQL, Blob Storage, Key Vault, Monitor) for scalable microservices deployment.
- Optimized relational databases (PostgreSQL, MySQL, Oracle, DB2) with expertise in query tuning, indexing, and stored procedure development.
- Managed semi-structured and distributed data using NoSQL databases such as MongoDB and DynamoDB.
- Implemented high-performance caching solutions with Redis to enhance application efficiency.
- Adopted Agile and Scrum methodologies, practicing TDD and BDD with testing frameworks like JUnit, Mockito, and Jest for robust code quality.
- Utilized monitoring and observability tools, including Prometheus, Grafana, ELK Stack, and Splunk, to ensure application performance and system health.
- Demonstrated strong proficiency in data structures, algorithms, debugging, and performance optimization for distributed systems.
- Streamlined development workflows using Git, Bitbucket, Jira, Confluence, and Maven for version control, project management, and build automation.

Certifications:

- Oracle Java SE 11 developer.
- Oracle Cloud Infrastructure 2025 Certified Developer Professional
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional
- Microsoft Solution Architecture
- Oracle Java SE 8 Programmer.
- AWS certified Developer.
- Microsoft certified Azure Fundamentals.
- Software Engineering Job Simulation by JP MORGAN Chase & Co

Technical Skills:

Front-End Development	React.js, Angular, Vue.js, JavaScript, TypeScript, CSS3, Sass, Bootstrap, AJAX, jQuery
Back-End Development	Java, Scala, Spring Boot, Spring MVC, Spring Security, Hibernate, Node.js, Express.js, Django, Flask, PHP, Laravel
APIs & Security	RESTful APIs, GraphQL, SOAP, FHIR, OAuth 2.0, JWT, API Gateway, Multi-Factor Authentication, JSON
Microservices & Messaging	Spring Boot Microservices Architecture, Kafka, RabbitMQ
Databases	MySQL, MongoDB, Oracle, Cassandra, SQL, PL/SQL, SQLite
CI/CD & Version Control	Jenkins, GitHub Actions, Git, GitLab, Bitbucket, CI/CD Pipelines
Testing & Monitoring	JUnit, Mockito, Cypress, Postman, ELK Stack (Elasticsearch, Logstash, Kibana), Prometheus, Grafana, Splunk, Cucumber
Cloud & DevOps	AWS (EC2, S3, Lambda, RDS, DynamoDB, IAM), Azure, Kubernetes, Docker, Terraform, CloudFormation, AWS CLI, GCP
Agile & Project Management	JIRA, Scrum, Kanban, SAFe, Sprint Planning, Documentation, Confluence
Soft Skills	Problem-solving, Time Management, Leadership, Communication, Decision-making, Documentation, Presentation

Professional Experience:

Client: Credit Union of Atlanta, Atlanta, GA

Project: Core Transformation & Cloud Modernization

Role: Sr. Software Engineer

Dec 2024-Present

Description: The Core Transformation & Cloud Modernization program is a mission-critical enterprise initiative that migrated legacy monolithic Java systems to cloud-native microservices on AWS/Azure, leveraging Spring Boot, Docker, and Kubernetes. It implemented event-driven Kafka pipelines and serverless functions, migrated Oracle databases to PostgreSQL/DynamoDB, and adopted Infrastructure as Code with Terraform and Helm—integrated into GitHub Actions CI/CD for automated testing, deployment, and monitoring.

Responsibilities:

- Designed and developed responsive front-end interfaces using Angular and TypeScript, optimizing member-facing digital banking platforms to deliver an enhanced user experience.
- Architected and implemented RESTful APIs with Spring Boot to support core banking services, ensuring seamless integration with legacy systems during core transformation.
- Spearheaded cloud migration initiatives, deploying microservices to AWS using Kubernetes, optimizing scalability and reliability for modernized banking applications.
- Deployed and maintained scalable backend applications on AWS EC2 instances, configuring security groups, key pairs, and auto-scaling policies for high availability.
- Developed event-driven microservices using AWS Lambda, triggered by S3 uploads, API Gateway events, and DynamoDB streams.
- Managed S3 buckets for static assets, logs, and intermediate data storage in ETL workflows with lifecycle policies for cost optimization.
- Optimized PostgreSQL database schemas to handle high-volume transactional data, improving query performance by 35% for core banking processes.
- Developed CI/CD pipelines using Jenkins and Docker, enabling automated testing and zero-downtime deployments for cloud-native applications.
- Designed interactive dashboards with Angular to visualize member data trends, supporting data-driven decisions for credit union operations.
- Monitored application performance using Prometheus and Grafana, reducing system downtime by 40% through proactive issue resolution in cloud environments.
- Enforced security standards with SonarQube and OWASP scans, ensuring compliance with credit union

regulations and achieving 100% audit readiness.

- Documented API contracts and system architectures in Confluence, facilitating cross-team collaboration and supporting core transformation audits.
- Participated in SAFe agile processes, leading sprint planning and demos to align development with Credit Union of Atlanta's strategic modernization goals.
- Mentored junior developers on Angular, Spring Boot, and agile methodologies, aligning with credit union coding standards to boost team productivity by 20%.
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Environment: Angular 17, TypeScript, Spring Boot, RESTful APIs, AWS (EKS), Kubernetes, PostgreSQL, Jenkins CI/CD, Docker, Prometheus, Grafana, SonarQube, OWASP, Confluence, SAFe Agile

Client: Centene Corporation, St. Louis, MO

Feb 2024 - Nov 2024

Project: Intelligent Health Data Exchange & Care-Insights Platform

Role: Sr Software Engineer

Description: The Intelligent Health Data Exchange & Care-Insights Platform is a mission-critical microservices solution built with Java/Spring Boot and Node.js that ingests, normalizes, and enriches HL7 and FHIR data via Kafka and Graph QL; integrates EHR, lab, wearable, and claims systems; and delivers unified, real-time analytics through a React portal, deployed on AWS EKS with Terraform and Helm.

Responsibilities:

- Designed and developed multi-tenant RESTful APIs using Spring Boot, adhering to Centene's API governance standards, ensuring scalability and compliance.
- Developed React-based front-end modules for care coordination workflows, enabling care managers to identify and flag high-risk members efficiently.
- Built interactive dashboards with React.js to visualize member enrolment, claims, and utilization trends, improving data accessibility by 30%.
- Styled and enhanced the look and feel of web pages using Sass, leveraging variables, mixins, and nesting to produce scalable, maintainable CSS structures.
Implemented and maintained dynamic client-side functionality with jQuery, writing and testing scripts to optimize application speed, scalability, and technical feasibility of UI/UX designs.
- Optimized Spring Boot APIs for high-throughput performance, reducing latency by 25% and improving reliability for mission-critical services.
- Architected and optimized PostgreSQL and MongoDB database schemas to manage 5TB+ of longitudinal health and claims data, enhancing query performance by 40%.
- Integrated FHIR R4 endpoints and rest API services for seamless connectivity with Centene's claims adjudication and eligibility systems.
- Implemented secure OAuth2 and OpenID Connect authentication flows using Okta, ensuring compliance with HIPAA and SOC 2 security standards.
- Wrote unit and integration tests using JUnit and Mockito, achieving 85% code coverage for back-end services, ensuring robust functionality.
- Maintained CI/CD pipelines in Jenkins to deploy Dockerized microservices on AWS EKS, enabling zero-downtime rollouts and streamlining deployments.
- Automated EC2 provisioning and environment setup using CloudFormation and Terraform, reducing manual deployment errors by 80%.
- Configured Kubernetes manifests (YAML) for deployments, services, and config maps, enabling environment-specific setups across dev, staging, and prod.
- Implemented secure S3 access using IAM roles/policies and pre-signed URLs to ensure compliance and data privacy.
- Monitored application health using Prometheus and Grafana, reducing mean time to resolution (MTTR) by 50% through proactive issue detection.
- Enforced HIPAA, SOC 2, and Centene's security standards using SonarQube and OWASP scans, achieving the compliance in security audits.
- Documented API contracts, data flows, and runbooks in Confluence, improving cross-team collaboration and audit readiness.

- Actively participated in SAFe agile processes, including sprint planning, demos, and retrospectives, driving 20% faster delivery cycles.
- Mentored junior developers on Java, Spring Boot, React, and Centene's coding standards, fostering adherence to agile best practices.

Environment: Java, Spring Boot, Spring MVC, Spring Security, Hibernate, React, jQuery, JavaScript, TypeScript, HTML5, CSS3, Sass, REST APIs, FHIR R4/R5, Docker, Kubernetes stable release, Jenkins, SonarQube, Apache Tomcat, JBoss EAP, CI/CD pipelines, Git, Windows & Linux environments, Agile methodology, SAFe framework

Client: Connet, Vancouver, WA

Sep 2023 – Feb 2024

Project: Deep Zoom CAD Visualization & Fleet Insights Platform

Role: Senior Software Engineer

Description: The Deep Zoom CAD Visualization & Fleet Insights Platform is a mission-critical microservices solution built on Java/Spring cloud and Node.js that delivers high-fidelity, interactive 3D CAD model rendering (via WebGL/Three.js) alongside real-time fleet telematics analytics through Kafka Streams. It integrates IoT device data and CAD repositories into Cassandra and Redis stores, exposed via a React-based portal and deployed on AWS EKS with Terraform and Helm for scalable, secure operations.

Responsibilities:

- Architected and implemented React/TypeScript front-end modules for seamless 3D CAD model visualization, rendering, and annotation within the Deep Zoom platform.
- Developed Spring Boot 3 microservices serving REST and GraphQL APIs to stream high-resolution CAD layers and metadata to web and mobile clients.
- Designed MongoDB schemas and Cassandra data models to store, index, and version-control massive multi-terabyte CAD file hierarchies and fleet analytics time-series.
- Integrated Kafka event streams to propagate real-time fleet telemetry (location, diagnostics, alerts) from IoT devices to Angular dashboards.
- Built OAuth 2.0 / JWT security layer with Spring Security to enforce role-based access and Multi-Factor Authentication for CAD intellectual-property compliance.
- Exposed secure, rate-limited GraphQL endpoints via API Gateway to unify CAD asset queries, asset-change subscriptions, and fleet performance analytics.
- Implemented reactive Node.js/Express services for lightweight geofencing and rule-engine processing of fleet events before persisting into Cassandra.
- Optimized MySQL transactional repository for configuration management, user and permission metadata while offloading heavy CAD geometry to Mongo Grid FS.
- Containerized all services with Docker and orchestrated via Kubernetes on AWS EKS, leveraging Terraform and CloudFormation for infrastructure-as-code.
- Integrated S3 with AWS Lambda, Glue, and Data Lake solutions, enabling automated ingestion and real-time processing of structured and unstructured data.
- Built real-time ETL pipelines using Lambda to process streaming data and feed it into analytics services.
- Established CI/CD workflows in GitHub Actions to achieve zero-downtime deployments, automated JUnit/Mockito test suites, and Cypress UI regression tests.
- Deployed Redis clusters on AWS Elasticache for low-latency caching of frequently accessed CAD thumbnails and fleet KPI snapshots.
- Instrumented microservices with Prometheus metrics, Grafana dashboards, ELK centralized logging, and Splunk alerts for proactive monitoring and SLA adherence.
- Led migration from RabbitMQ to Kafka for high-throughput telemetry ingestion, reducing end-to-end message latency and ensuring exactly once semantics.
- Coordinated Agile Scrum ceremonies in JIRA, breaking epics into executable user stories, refining acceptance criteria, and delivering incremental features per sprint cadence.

Environment: React, TypeScript, WebGL / Three.js, Spring Boot, Node.js, Express.js, REST, GraphQL, MongoDB, Cassandra, MySQL, Kafka Streams, OAuth 2.0 / JWT, Docker, Kubernetes (AWS EKS stable 1.24–1.26), Terraform,

CloudFormation, GitHub Actions, CI/CD, JUnit, Mockito, Cypress, Redis (Elastic Cache), Prometheus, Grafana, ELK Stack, Splunk, GitLab, JIRA, Agile Scrum

Client: HDFC Bank, Mumbai, India

Jul 2019 – Aug 2023

Role: Software Engineer

Project: Real-Time Transaction Fraud Monitoring Platform

Description: Developed a mission-critical real-time transaction fraud monitoring platform using Java Spring Boot microservices and Kafka Streams, deployed on Azure Kubernetes Service (AKS). The system processes live transaction data for both rule-based and ML-driven fraud detection. It integrates with payment gateways, core banking services, and real-time ML scoring APIs. Analysts interface with the system via a React portal. The infrastructure was provisioned and managed using Terraform and Helm, with Azure-native services ensuring compliance, scalability, and security.

Responsibilities:

- Architected and implemented React 18 dashboards for real-time fraud monitoring using WebSocket and Server Sent Events.
- Developed Spring Boot 3 microservices with REST and GraphQL APIs to ingest, score, and annotate card and ACH transactions.
- Orchestrated Kafka clusters for exactly-once, fault-tolerant event streaming into downstream risk engines.
- Designed and tuned Cassandra schemas for high throughput writes and efficient time range queries on transaction lineage.
- Secured applications with OAuth 2.0, JWT, and multi-factor authentication in compliance with PCI DSS.
- Optimized performance by configuring Redis in-memory caching of statistical risk models and customer profiles.
- Containerized microservices with Docker and deployed to Azure Kubernetes Service (AKS) using Terraform, standardizing environments.
- Built CI/CD pipelines in Jenkins, orchestrating JUnit, Mockito, Postman, and Cypress test suites.
- Instrumented systems with Prometheus, Grafana, ELK Stack, and Splunk for end-to-end monitoring and anomaly detection.
- Managed Azure Database for Oracle (or hosted Oracle DB on Azure VMs) with Spring Data JPA for relational metadata and offloaded logs to MongoDB (via Cosmos DB Mongo API if on Azure).
- Exposed RESTful webhook callbacks to partner banks and networks for synchronous block/allow decisions.
- Implemented a Node.js/Express API Gateway for rate limiting, schema validation, and Azure Web Application Firewall (WAF) integration.
- Architected and optimized SQLite database schemas and queries for embedded application storage, ensuring efficient local data retrieval and data integrity in resource-constrained environments.
- Involved in end-to-end web development projects, designing, coding, and modifying websites according to client specifications using modern frontend and backend technologies.
- Designed and deployed serverless applications using Azure Functions, API Management, and related Azure services, reducing operational costs by 40% and improving system scalability by 60%.

Environment: Java 11, Spring Boot, Node.js LTS chain: Node.js, Express.js, PHP Laravel, REST APIs, Azure Functions, API Management, SQLite, MySQL, JavaScript, Sass, jQuery, Docker, Kubernetes via Azure AKS, Jenkins, ELK Stack, Prometheus, Grafana, Postman, Cypress, JUnit, Windows & Linux environments, Agile Scrum.

Client: Apollo Hospitals Enterprise Limited, Chennai, India

Jan 2017 – Jun 2019

Project: Clinical Intelligence Engine (CIE) & Connected Care Platform

Role: Associate Software Engineer

Description: The Clinical Intelligence Engine and Connected Care Platform was a high-availability, clinician-facing healthcare solution designed to ingest real-time patient vitals and EHR updates to deliver actionable alerts. Built on a microservices architecture using Java/Spring Boot and deployed to Google Kubernetes Engine (GKE), the

platform integrated Epic FHIR, MongoDB Atlas ML pipelines, and legacy SOAP services behind a unified GraphQL API Gateway. Clinician-facing dashboards were built in React, with infrastructure managed using Terraform and Helm, and deployed on Google Cloud Platform for scalability and security.

Responsibilities:

- Developed and maintained responsive React components and single page applications to support clinician workflows.
- Designed and implemented RESTful APIs using Spring Boot, handling patient data retrieval and updates.
- Collaborated with product owners and UX designers in agile ceremonies to refine requirements and plan two-week sprints.
- Configured and deployed microservices to Google Kubernetes Engine (GKE) using Docker, Helm charts, and terraform for consistent environments.
- Modelled and managed Cassandra key spaces for time series vital data, applying TTL rules and indexes for efficient queries.
- Integrated Redis caching layers (via Google Cloud Memory store) to accelerate repeat queries and store session state, reducing backend load.
- Wrote JUnit and Mockito unit tests, achieving high code coverage and catching regressions early in GitHub Actions CI pipelines.
- Instrumented Prometheus metrics and created Grafana dashboards to monitor service health, latency, and error rates.
- Implemented Spring Security with RBAC and ABAC policies to enforce user permissions and protect PHI on every request.
- Orchestrated Kafka Streams jobs (via Confluent Cloud on GCP) to process real-time vitals and alert events, ensuring reliable data delivery.
- Troubleshoot production incidents, conducted root cause analysis, and applied hotfixes without impacting uptime.
- Authored technical documentation and API specs in Confluence and Swagger to support team onboarding and knowledge sharing.

Environment: React, Java, Spring Boot, Spring Security (RBAC/ABAC) , RESTful APIs, Docker , Helm 2.0, Terraform , GCP GKE, Cassandra , Redis , Kafka Streams , JUnit , Mockito , Cypress , Prometheus , Grafana, Confluence, Swagger.